

Linear Birth–Death Branching at Every Nonnegative Rate Pair: An Exact Finite-Time Möbius and Transition Theorem

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Abstract

For the continuous-time branching chain with per-particle birth and death rates $\lambda, \mu \geq 0$, we prove one formula package valid for every initial population $z \in \mathbb{N}_0$. The one-ancestor probability generating function is a Möbius semigroup. Off criticality its clock is $\delta = e^{-(\lambda-\mu)t}$; at criticality the nonsingular clock is $\tau = \lambda t$. Its zero-modified geometric expansion yields the exact z -ancestor transition as a binomial mixture over surviving ancestral lineages, with a conditional negative-binomial sum. This prevents the common, nonuniform replacement by one negative-binomial law. Differentiation gives the mean and variance, including all pure and zero-rate boundaries.

Keywords: Markov branching process; linear birth–death chain; probability generating function; quasi-stationarity; Yaglom limit; martingale limit.

中文摘要

本文对任意非负出生率、死亡率及任意非负整数初值，给出线性出生死亡分枝过程的完整公式。单祖先概率生成函数构成 Möbius 半群；任意初值的转移律是幸存祖先数的二项混合，给定幸存祖先数后才是负二项和，不能在全参数范围内统一写成一个负二项分布。全部纯出生、纯死亡、零速率、零初值和零时刻边界均被显式闭合。

关键词：出生死亡过程；分枝过程；概率生成函数；准平稳分布；鞅极限。

1 Source lock and model

Let $Z = (Z_t)_{t \geq 0}$ be the minimal continuous-time Markov chain on $\mathbb{N}_0$ with generator

$$(Gf)(n) = \lambda n\{f(n+1) - f(n)\} + \mu n\{f(n-1) - f(n)\}, \quad \lambda, \mu \geq 0. \quad (1)$$

State zero is absorbing and $Z_0 = z \in \mathbb{N}_0$. A Yule process with rate λ dominates the number of births, so the chain is nonexplosive at every finite time. Distinct initial particles found independent descendant families. Consequently, if

$$F_t(s) = \mathbb{E}_1[s^{Z_t}], \quad 0 \leq s \leq 1,$$

then $\mathbb{E}_z[s^{Z_t}] = F_t(s)^z$. This precise source family and its linear growth terminology are locked to the classical report of Karlin and McGregor [1]; no priority claim is made here.

2 The finite-time theorem

Put $r = \lambda - \mu$. For $r \neq 0$ define $\delta = e^{-rt}$; for $r = 0$ put $\tau = \lambda t$.

Theorem 1 (Möbius PGF semigroup). For all $\lambda, \mu \geq 0$ and $t \geq 0$,

$$r \neq 0: \quad F_t(s) = \frac{\mu(1-s) - \delta(\mu - \lambda s)}{\lambda(1-s) - \delta(\mu - \lambda s)}, \quad (2)$$

$$r = 0: \quad F_t(s) = \frac{\tau + (1-\tau)s}{1 + \tau - \tau s}. \quad (3)$$

The second line includes $\lambda = \mu = 0$, when $\tau = 0$ and $F_t(s) = s$. Moreover $F_t \circ F_u = F_{t+u}$.

Proof. Conditioning over the first event of one particle gives the backward equation

$$\partial_t F_t = \lambda(F_t^2 - F_t) + \mu(1 - F_t) = (F_t - 1)(\lambda F_t - \mu), \quad F_0(s) = s. \quad (4)$$

Separating the two roots gives (2); when the roots coalesce, integrating $\partial_t F = \lambda(F - 1)^2$ gives (3). The formulas also follow from the matrices, acting by $(As + B)/(Cs + D)$,

$$M_\delta = \frac{1}{r} \begin{pmatrix} \delta\lambda - \mu & \mu(1-\delta) \\ \lambda(\delta-1) & \lambda - \delta\mu \end{pmatrix}, \quad N_\tau = \begin{pmatrix} 1-\tau & \tau \\ -\tau & 1+\tau \end{pmatrix}.$$

Direct multiplication yields $M_{\delta_1} M_{\delta_2} = M_{\delta_1 \delta_2}$ and $N_{\tau_1} N_{\tau_2} = N_{\tau_1 + \tau_2}$; both matrices are the identity at zero time. Since $\delta(t+u) = \delta(t)\delta(u)$ and $\tau(t+u) = \tau(t) + \tau(u)$, the semigroup claim follows. $\square$

Theorem 2 (one ancestor and every- z transition). Define, for $r \neq 0$,

$$p_0 = \frac{\mu(1-\delta)}{\lambda - \mu\delta}, \quad \beta = \frac{\lambda(1-\delta)}{\lambda - \mu\delta}, \quad (5)$$

and, for $r = 0$, define

$$p_0 = \beta = \frac{\tau}{1 + \tau}. \quad (6)$$

Then $0 \leq p_0 \leq 1$, $0 \leq \beta < 1$, and

$$\mathbb{P}_1(Z_t = 0) = p_0, \quad \mathbb{P}_1(Z_t = n) = (1 - p_0)(1 - \beta)\beta^{n-1}, \quad n \geq 1. \quad (7)$$

For every $z \in \mathbb{N}_0$,

$$\mathbb{P}_z(Z_t = 0) = p_0^z, \quad (8)$$

$$\mathbb{P}_z(Z_t = n) = \sum_{k=1}^{\min(z,n)} \binom{z}{k} p_0^{z-k} (1 - p_0)^k \binom{n-1}{k-1} (1 - \beta)^k \beta^{n-k}, \quad n \geq 1. \quad (9)$$

Proof. Algebra rewrites either PGF in Theorem 1 as

$$F_t(s) = p_0 + (1 - p_0) \frac{(1 - \beta)s}{1 - \beta s}. \quad (10)$$

For $r > 0$, $0 < \delta \leq 1$; for $r < 0$, $\delta \geq 1$. These two sign cases in (5), together with the pure-rate edges, give the claimed parameter ranges. Expanding the geometric denominator proves (7).

For z independent ancestral lines, let K be the number nonextinct at time t . Then $K \sim \text{Bin}(z, 1 - p_0)$. Given $K = k$, the k positive family sizes are independent geometric variables on $\{1, 2, \dots\}$ with success $1 - \beta$; their sum has probability $\binom{n-1}{k-1} (1 - \beta)^k \beta^{n-k}$. Mixing over K gives (8)–(9). $\square$

Why this is not uniformly one negative-binomial law. The mixing shape K is random whenever several ancestral lines can independently die. Thus the all-parameter law must retain the binomial survivor mixture; only after conditioning on $K = k$ is the positive sum one negative-binomial law with fixed shape k . At special parameter values the mixture can collapse—for example, if $p_0 + \beta = 1$, then $F_t(s)^z$ itself has an ordinary negative-binomial form. This exception is exactly why the claim is about the uniform formula rather than an impossible pointwise ban.

Corollary 3 (moments). For every $z \in \mathbb{N}_0$,

$$\mathbb{E}_z Z_t = z e^{rt}, \quad (11)$$

$$\text{Var}_z Z_t = \begin{cases} z \frac{\lambda + \mu}{r} e^{rt} (e^{rt} - 1), & r \neq 0, \\ 2z\lambda t, & r = 0. \end{cases} \quad (12)$$

Proof. Differentiate $F_t(s)^z$ at $s = 1$. From (10), one ancestral family has mean $(1 - p_0)/(1 - \beta)$ and second moment $(1 - p_0)(1 + \beta)/(1 - \beta)^2$. Substitution of (5) or (6), followed by addition across independent ancestors, gives (11)–(12). $\square$

All finite-time degeneracies. The formulas are literal at every boundary:

Boundary	One-ancestor parameters	Result for z ancestors
$t = 0$	$p_0 = \beta = 0$	$Z_t = z$
$z = 0$	any admissible pair	$Z_t = 0$
$\lambda = \mu = 0$	$p_0 = \beta = 0$ for all t	$Z_t = z$
$\mu = 0 < \lambda$	$p_0 = 0, \beta = 1 - e^{-\lambda t}$	shifted negative-binomial sum
$\lambda = 0 < \mu$	$p_0 = 1 - e^{-\mu t}, \beta = 0$	$\text{Bin}(z, e^{-\mu t})$
$\lambda = \mu > 0$	$p_0 = \beta = \tau/(1 + \tau)$	critical formula (9)

Revision focus. Round 0 closes the Möbius semigroup, exact survivor-mixture transition, moments and every finite-time degeneracy.

Declarations

Data and code availability. Package-local synthetic exact evidence and deterministic code accompany the paper. No observational population data are used.

Ethics. No human, animal, personal, clinical or sensitive data are used; approval is not applicable.

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References

- [1] S. Karlin and J. McGregor, “Linear Growth, Birth and Death Processes,” Technical Report KAR ONR 3, Stanford University, January 1958. <https://statistics.stanford.edu/technical-reports/linear-growth-birth-and-death-processes>; persistent record <https://purl.stanford.edu/fx071vs8733>.